

THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DARES SALAAM REGION

FORM SIX MOCK EXAMINATION

CHEMISTRY 2

GEPAM science hub

WhatsApp 0748978398

Code: 132/2

Time: 3:00 Hours

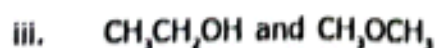
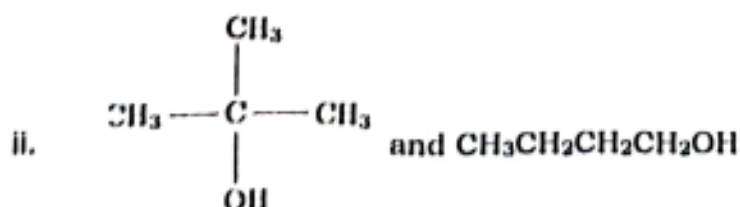
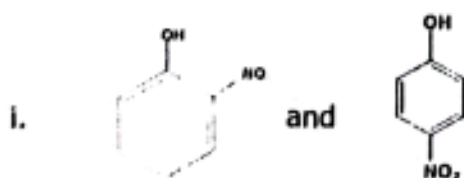
Tuesday 11th February, 2024 A.M

INSTRUCTIONS

1. This paper consists of Six (6) questions.
2. Answer only Five (5) questions.
3. Each question carries equal marks (20 marks).
4. Non programmable calculator may be used but cannot be shared.
5. Write your Examination Number on every page of your answer booklet(s).
6. The following constant may be useful.
 - Molar gas constant, $R = 8.314 JK^{-1} mol^{-1}$ or $0.082 atm mol^{-1} K^{-1} dm^3$
 - 1 Faraday = $96500 C mol^{-1}$ • $GMV = 22.4 dm^3$
 - 1 Litre = $1 dm^3 = 1000 cm^3$
 - Relative atomic masses:
 $Ar = 40, H = 1, O = 16, N = 14, Na = 23, K = 39, Pb = 207, I = 127$

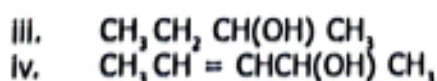
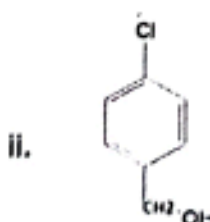
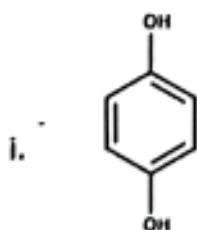
This paper consists of 4 printed pages

1. (a) (i) What are the three (3) conditions necessary for the partition coefficient of a solute between the two given solvents to remain constant?
 - (ii) If the partition coefficient of Q between ethoxyethane and water is 5.0, calculate mass of Q extracted from water, when a solution containing 10.0g of Q in 500cm³ of water is extracted with 100cm³ of ethoxyethane **(07 marks)**
 - (b) Explain why the pure components of the azeotropic mixture cannot be separated by means of fractional distillation. **(04 marks)**
 - (c) The azeotropic mixture between water and nitric acid (HNO₃) boils at 121°C and the boiling point of water and HNO₃ is 100°C and 78°C respectively. The azeotropic mixture composition is 68% HNO₃ and 32% H₂O.
 - (i) By using graph identify, the component which will be collected as a distillate when the above mixture with 50% composition of HNO₃ undergoes fractional distillation.
 - (ii) Draw a graph of vapor pressure against composition of the above water and nitric acid solution.
 - (iii) Identify the type of non-ideal solution exhibited by the mixture above. **(09 marks)**
2. a) Explain the following terms;
 - i) Electrochemical cell
 - ii) Standard hydrogen electrode
 - iii) Electrode potential **(03 marks)**
 - b) i) Calculate the equilibrium constant of cell. Given that $AP^{+}/Al, E^{\theta} = -0.259v$ and $Pb^{2+}/Pb, = -0.26v$
 - ii. Calculate the solubility product of Lead (II) Iodide if its solubility in water at 50 °C is 0.11g/dm³. What is its solubility in 0.01M sodium Iodide? Comment on the differences of solubility of Lead Iodide in the two given solutions. **(06 marks)**
 - c) Show that in case of first order reaction, the time required for 99.9% of the reaction to take place is ten times that required for half of the reaction **(05 marks)**
 - d) The rate of a reactions triples when temperature changes from 20 °C to 50 °C. Calculate the activation energy of a given reaction. **(06 marks)**
 3. (a) The members if the following pairs of isometric compounds have different boiling points. Indicate which member have high boiling point and suggest reason for the differences.



(03 marks)

- (b) Explain the following terms as used in organic chemistry
- Polymers
 - Thermosetting polymer
 - Thermoplastic polymer
- (03 marks)
- c) Write the polymers molecules that will be formed from the following monomers
- $\text{H}_2\text{NCH}(\text{CH}_3)\text{COOH}$
 - $\text{H}_2\text{N} - (\text{CH}_2)_6 - \text{NH}_2$
 - $\text{H} - \text{C} \equiv \text{C} - \text{H}$
 - $\text{HOOC} - (\text{CH}_2)_4 - \text{COOH}$
- (08 marks)
- d) Write the formulae of the following complex compounds;
- Tetraamminechloronitrocobalt (III) chloride
 - Potassium tetracyanonickellate (II)
 - Triamminebromoplatinum (II) chloride
 - Potassium hexafluorocobaltate (III)
- (06marks)
4. a) Account for the following
- Ethylamine is soluble in water whereas aniline is not
 - Methylamine in water reacts with ferric chloride to precipitate hydrated ferric oxide
 - The boiling point of aliphatic amines is lower than those of the corresponding alcohols
- (06 marks)
- b) A compound X having molecular formula $\text{C}_3\text{H}_7\text{NO}$ reacts with bromine in the presence of NaOH to give another compound Y. The compound Y reacts with nitric (III) acid to form ethanol and nitrogen gas. Identify the compounds X and Y and write the chemical equations of the reactions involved. (04 marks).
- c) How will you bring about these conversions?
- 4- Nitrobenzaldehyde from benzene
 - Benzamide from nitrobenzene
 - m - bromiodobenzene from benzene
- (06 marks)
- d) By using IUPAC system name, the following compound



(04 marks)

5. (a) With one example explain the following terms.
- Bidentate ligands
 - Coordination number
 - Ambidentate ligands
 - Polydentate ligand
- (04 marks)**
- b) Briefly explain the following
- A concentrated aqueous copper(II) chloride solution is bright green in colour but changes to light blue when water added.
 - Hydrated copper (II) sulphate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) is blue in colour but when heated it becomes white.
 - Transition elements have higher enthalpies of atomization.
- (03 marks)**
- c) A cobalt complex having molecular formula of $\text{Co}(\text{NO}_2)_2\text{Cl}_4\text{NH}_3$ give an electrical conductance equivalent to two ions per formula unit. When excess AgNO_3 solution is added it precipitate one mole of AgCl per formula unit.
- Write the formula of the complex
 - Give the IUPAC name of the complex
- (04 marks)**
- d) Briefly explain the extraction of aluminium from bauxite (assume silicon dioxide as main impurity)
- (09 marks)**
6. a) An unknown ester, E was found to have molecular formula $\text{C}_3\text{H}_8\text{O}_2$. When E was hydrolyzed with water in the presence of mineral acid, it produced carboxylic acid A and alcohol B. The treatment of B with SOCl_2 gave an alkyl chloride C. When C is treated with KCN a product D was formed which on hydrolysis with water produced carboxylic acid A.
- Deduce the structure and name of the original ester.
 - Identify A, B, C and D and write the chemical equations for the reactions involved
- (08 marks)**
- b) A carbonyl compound A was reduced with LiAlH_4 to give a compound B. B was dehydrated with conc. H_2SO_4 to give one product C with molecular formula C_5H_{10} . Ozonolysis of C gave ethanal and compound D ($\text{C}_3\text{H}_6\text{O}$). D gave positive iodoform test. Identify A, B, C and D giving their names and structural formula. Explain how you have deduced their structural formula and give equations for all above reactions.
- (08 marks)**
- c) Using your knowledge of particular function group of various types of organic compounds predict the products of the following reactions



- A mixture of iodine and NaOH
- Ethanoyl chloride, AlCl_3
- H_2 and Ni at 200°C
- Acidified $\text{K}_2\text{Cr}_2\text{O}_7$ at only $60-80^\circ\text{C}$

(04 Marks)

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- (b) (i) — The solution should be stable.
 — The solute should be non electrolyte.
 — The temperature should be constant.

(ii) $W_0 = 10g$.

$V_L = 100cm^3$

$V_L = 500cm^3$

$k_d = 5$

$W_v = ?$
 $n = 1$

$$W_v = \left(\frac{V_L}{V_L + V_L k_d} \right)^n W_0$$

$$W_v = \left(\frac{500}{500 + (5 \times 100)} \right) \times 10$$

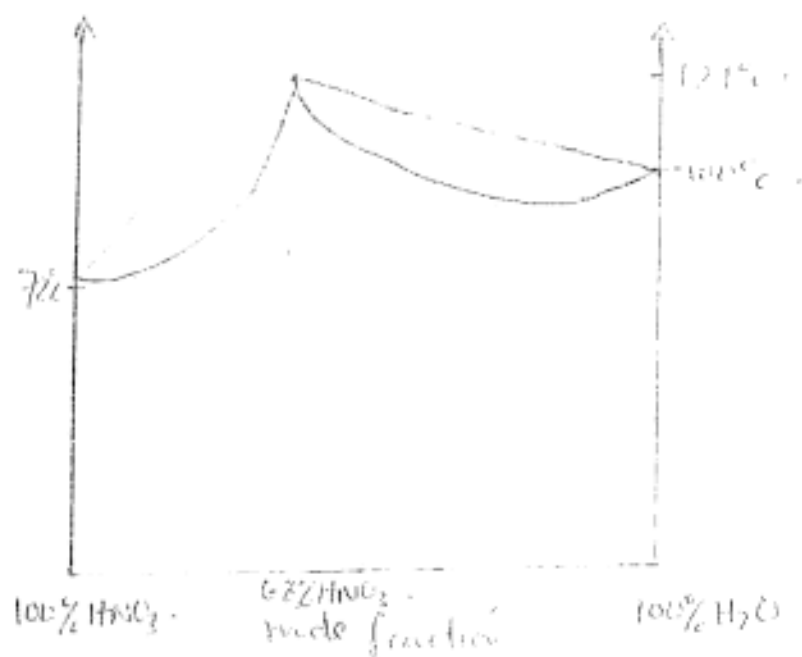
$W_v = 5g$

$W_{\text{extracted}} = W_0 - W_v = 10 - 5 = 5g$

∴ Mass of G extracted is 5g.

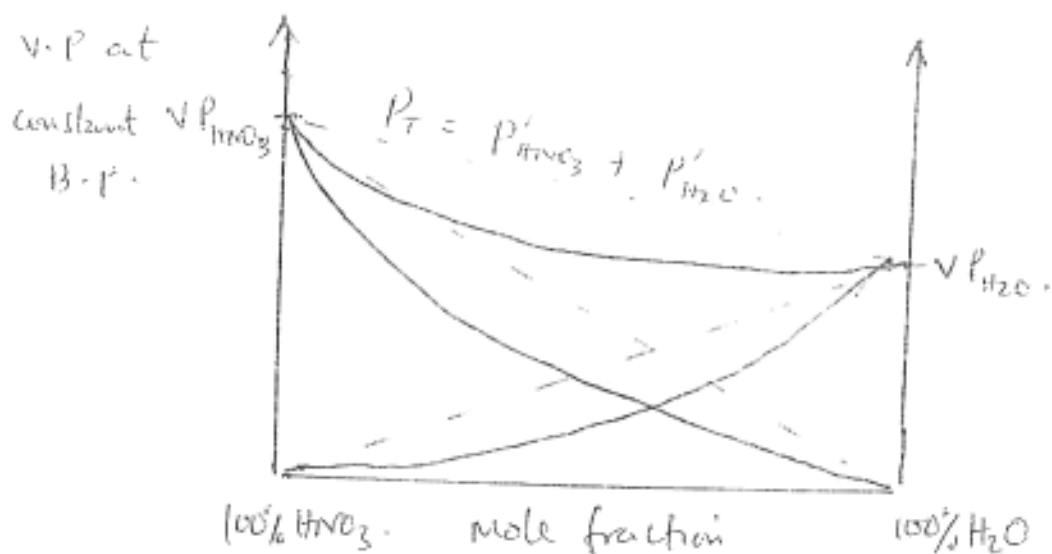
- (b) Azeotropic mixture can not be separated by fractional distillation because azeotrope is a constant boiling mixture. When an azeotropic mixture is boiled, the composition of vapour is the same as that of an unboiled mixture.

(c) (i)



in fractional distillation of 50% HNO_3
 the distillate will contain more water
 than nitric acid whereas the
 residue contain more HNO_3 than H_2O .
 in continuation of boiling and
 condensation, the distillate is pure H_2O
 and the residue is azeotropic mixture.

(ii) Vapour pressure against mole fraction.



(iii) Negative deviation.

2. (a) (i) Electrochemical cell

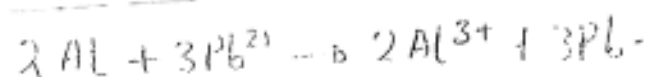
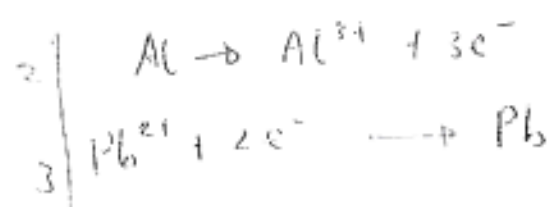
(ii) Standard hydrogen electrode

(iii) Electrode potential

(b) (i) $E_{\text{cell}} = E_R - E_S = E_{\text{Pb}}^{\ominus} - E_{\text{Al}}^{\ominus}$

$$E_{\text{cell}} = -0.126 - (-0.289) = 0.133 \text{ V}$$

$$K_c = \ln^{-1} \left(\frac{n E_{\text{cell}}}{0.0591} \right)$$



$$n = 6$$

$$K_c = \ln^{-1} \left(\frac{6 \times 0.133}{0.0591} \right)$$

$$K_c = 7.31 \times 10^5$$

(ii) molar mass of $\text{PbI}_2 = 207 + (127 \times 2) = 461 \text{ g mol}^{-1}$

$$\text{molar conc of } \text{PbI}_2 = \frac{\text{conc. in g dm}^{-3}}{\text{molar mass}}$$

$$= \frac{0.11 \text{ g dm}^{-3}}{461 \text{ g mol}^{-1}}$$

$$= 2.38 \times 10^{-4} \text{ mol dm}^{-3}$$

Molar conc. of PbI_2 is $2.59 \times 10^{-4} \text{ mol/dm}^3$

Let solubility of PbI_2 be x .



$$[\text{Pb}^{2+}] = x$$

$$[\text{I}^-] = 2x$$

$$K_{sp} = [\text{Pb}^{2+}][\text{I}^-]^2$$

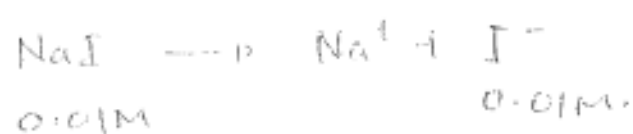
$$K_{sp} = (x)(2x)^2 = 4x^3$$

$$K_{sp} = 4(2.59 \times 10^{-4})^3$$

$$K_{sp} = 5.461 \times 10^{-11} \text{ mol}^3 \text{dm}^{-3}$$

$\therefore$ Solubility product of PbI_2 is $5.461 \times 10^{-11} \text{ mol}^3 \text{dm}^{-3}$.

In 0.01M NaI



$$[\text{Pb}^{2+}] = x$$

$$[\text{I}^-] = 2x + 0.01 \approx 0.01 \text{M}$$

$$K_{sp} = [\text{Pb}^{2+}][\text{I}^-]^2$$

$$5.461 \times 10^{-11} = (x)(0.01)^2$$

$$x = \frac{5.461 \times 10^{-11}}{(0.01)^2} = 5.461 \times 10^{-7} \text{M}$$

$\therefore$ Solubility of PbI_2 in 0.01M NaI is $5.461 \times 10^{-7} \text{M}$.

PbI_2 is less soluble in 0.01M NaI due to common ion effects.

$$2. \quad (c) \quad [A]_0 = 100\%$$

$$[A] = 100 - 99 = 1\%$$

from 1st order reaction

$$\ln \frac{[A]_0}{[A]} = kt$$

$$k = \left(\ln \frac{[A]_0}{[A]} \right) \cdot \frac{1}{t}$$

$$= \left(\ln \frac{100}{1} \right) \cdot \frac{1}{100}$$

$$\ln k_2 - \ln k_1 = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$3.1 - 2.3 = \frac{E_a}{8.314} \left(\frac{1}{298} - \frac{1}{323} \right)$$

$$0.8 = \frac{E_a}{8.314} \left(\frac{1}{298} - \frac{1}{323} \right)$$

$$E_a = 22000 \text{ J}$$

$$\ln \frac{k_2}{k_1} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\ln \frac{2}{1} = \frac{E_a}{8.314} \left(\frac{1}{298} - \frac{1}{T_2} \right)$$

$$\ln(2) = \frac{E_a}{8.314} \left(\frac{298 - T_2}{298 T_2} \right)$$

$$\ln(2) \times 8.314 \times 298 \times T_2 = E_a (298 - T_2)$$

$$22000 \text{ J} = 60.7 T_2$$

∴ Activation energy of the

reaction is 22000 J (22 kJ).

3. (a) (i)  because it can form intermolecular hydrogen bonds.

(ii) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ because it can form strong hydrogen bond.

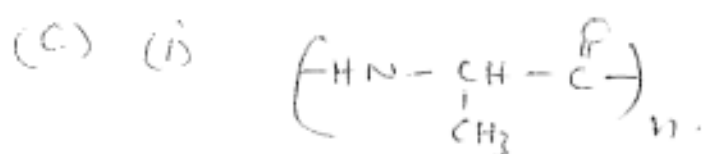
(iii) $\text{CH}_3\text{CH}_2\text{OH}$ b/c it can form hydrogen bond.

(b) (i) Polymers - Is a class of natural or synthetic substances composed of very large molecules made up of small chemical units called monomers.

(ii) Thermosetting polymer - Is a polymer that is obtained by irreversibly hardening a soft solid or viscous liquid polymer.

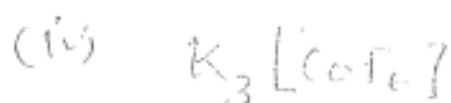
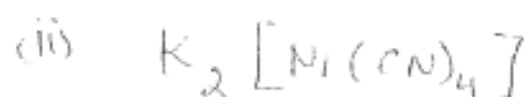
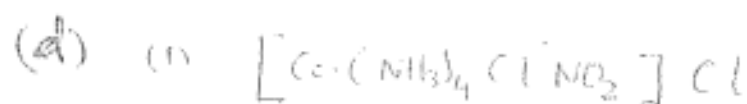
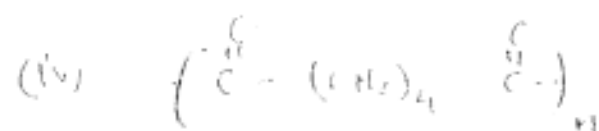
(iii) Thermoplastic polymer - Is any plastic material that becomes pliable or moldable at a certain elevated temperature and solidifies upon cooling.

(iv) Monomers - Are small chemical substances which combine to form large molecules/polymers.



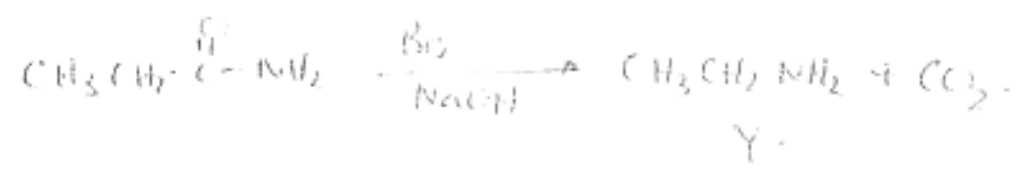
(ii)



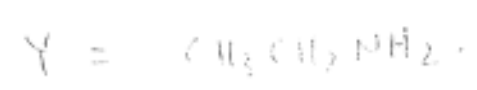
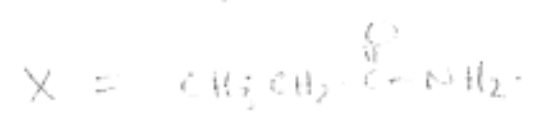


4. (a) (i) Ethylamine is soluble in water due to its ability to form hydrogen bonds with water. On the other hand, methylamine is hydrophobic due to presence of bent bond hence do not form hydrogen bonds.
- (ii) Methylamine in water react with ferric chloride to precipitate hydrated ferric oxide due to +I-effect of methyl group. CH_3NH_2 is more basic than water. Therefore in water methylamine produces OH^- ions by accepting H^+ ion from water. Ferric chloride (FeCl_3) dissociates in water to form Fe^{3+} and Cl^- ions.
- (iii) Alcohols form strong hydrogen bonds in comparison to aliphatic amines due to high electronegativity of oxygen than nitrogen.

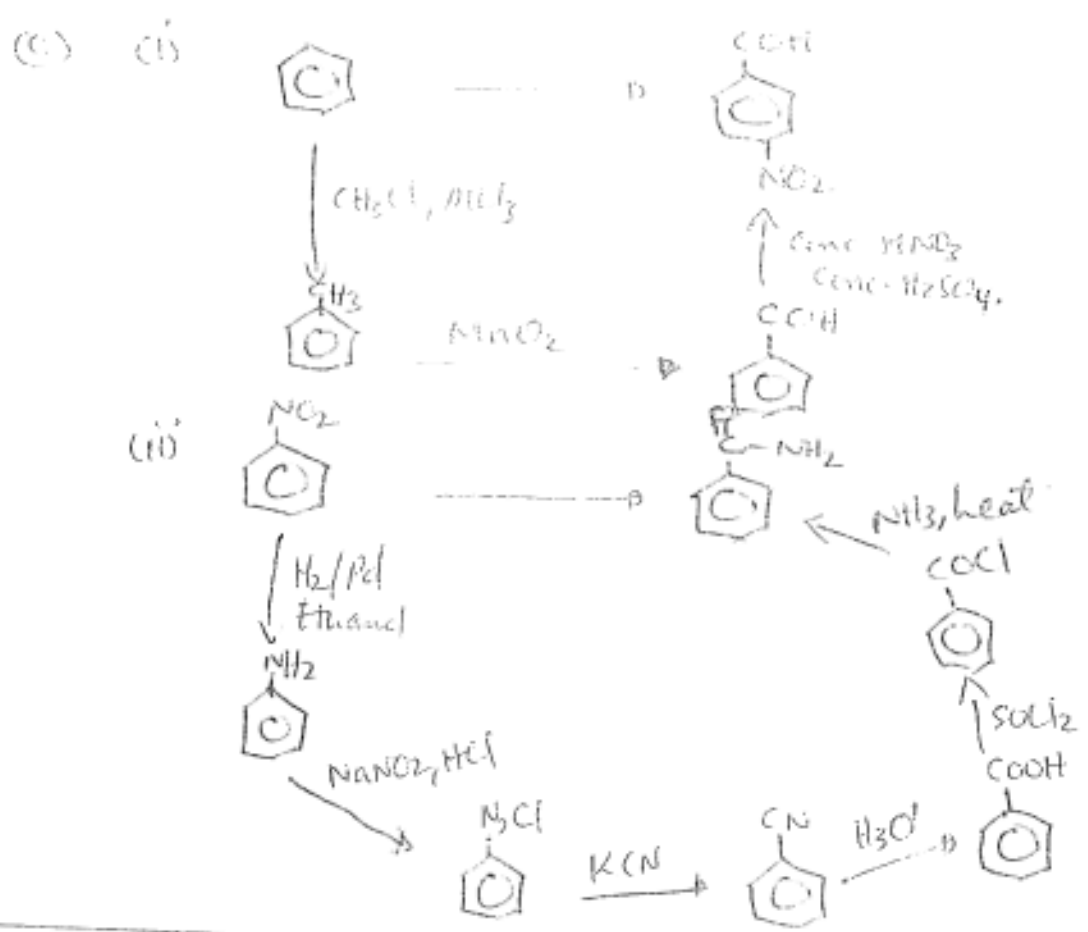
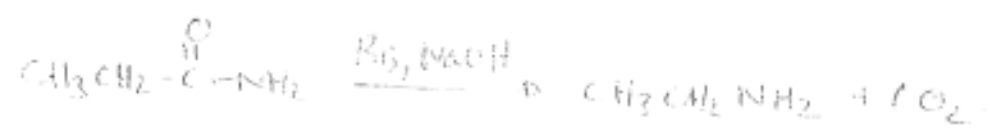
4. (b) A compound X is an amide b/c it reacts with Br_2 in presence of NaOH .
 $\therefore$ f of X is $\text{CH}_3\text{CH}_2\text{C}(=\text{O})\text{NH}_2$

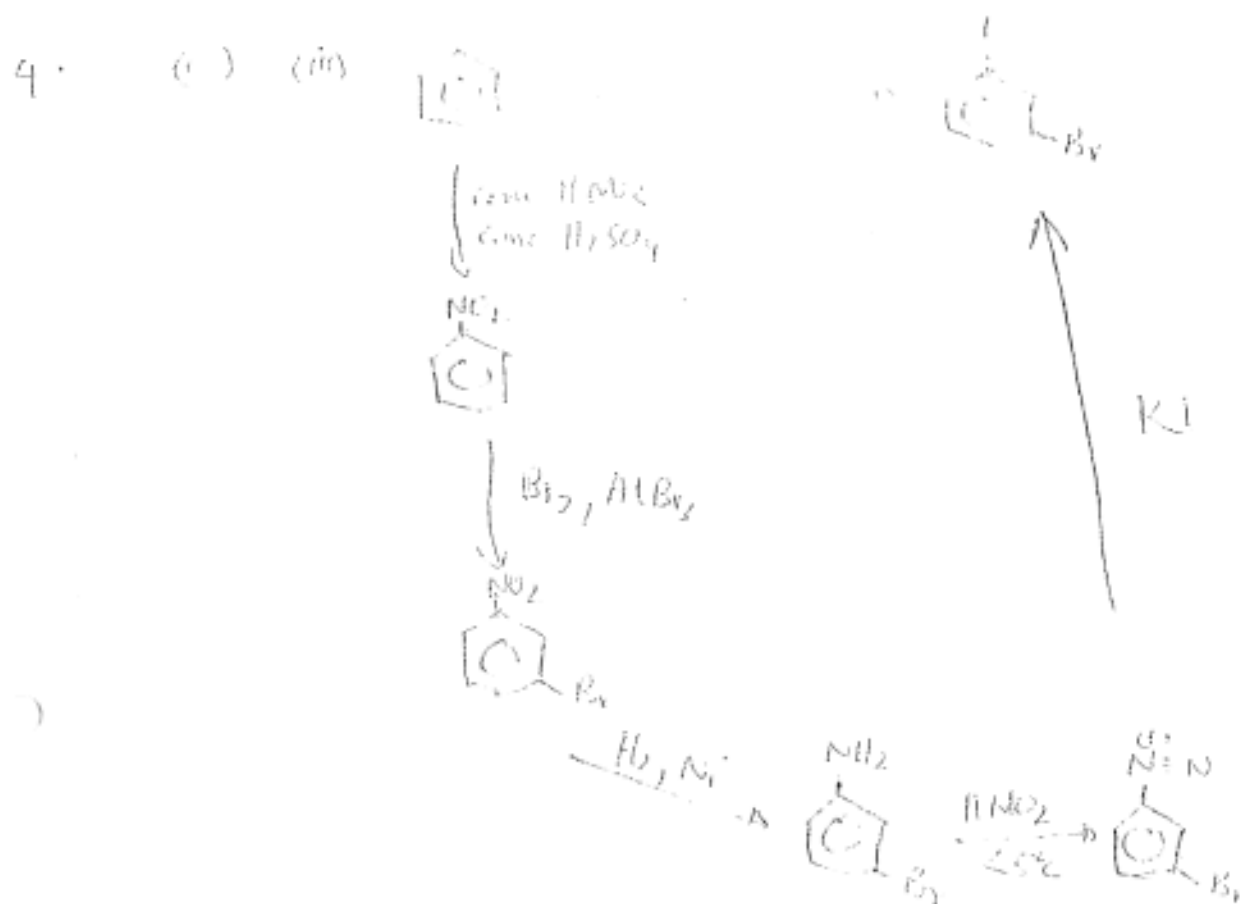


The s.f of



Chemical name Name.





(c) (i) 4-Hydroxyphenol

(ii) 4-chlorophenylmethanol

(iii) Butan-2-ol

(iv) Pent-3-en-2-ol.

5. (a) (i) Bidentate ligands - Are ligands which can form two dative bonds with the central metal atom.
E.g. ethylenediamine (en).

(ii) Coordination number - Is the number of dative bonds formed between ligands and central metal ion.
E.g. $[\text{CoF}_6]^{3-}$.

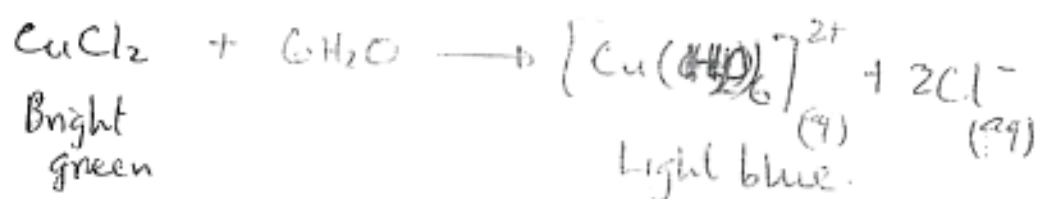
- 5 (iii) Ambidentate ligands: Are ligands with two donor atoms but use one donor atom at once.



- (iv) Polydentate ligands: Are ligands with many donor atoms.



- (b) (i) The colour change of concentrated aqueous copper (II) chloride from bright green to light blue when diluted is due to change in the electronic environment of the copper ion. A concentrated solution results in green colour due to lesser d-d transitions, while dilution causes an increase in the d-d transitions and results in a light blue colour.

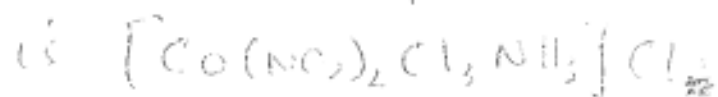


- (ii) Hydrated copper (II) sulphate have blue colour due to presence of water ligands which split the d-orbital and lead to light absorption. The anhydrous copper (II) sulphate is white due to absence of water ligands.

(b) (iii) This is because of large number of unpaired electrons in their atoms, transition elements have stronger interatomic interaction and hence stronger bonding between atoms resulting in higher enthalpies of atomization.

(c) (i) Since the complex form one mole of AgCl with AgNO_3 hence the complex has only one Cl outside the complex.

Therefore the complex formula



(ii) Ammine trichlorodinitrocobalt (III) Chloride

(d) $\text{Al}(\text{OH})_3$

$\text{Al}(\text{OH})_3$ is amphoteric with NaOH and HCl as follows



The above gets reactions to show that $\text{Al}(\text{OH})_3$ is amphoteric. It reacts with NaOH to form $[\text{Al}(\text{OH})_4]^-$ and with HCl to form AlCl_3 .
 i.e. hydrolysis AlN gives $\text{Al}(\text{OH})_3$.



the reaction of the pure substance with
 water is as follows:



On heating, the pure substance

forms a white solid which is called as

phosphorus pentoxide and phosphorus pentoxide

and phosphorus pentoxide is a white solid

which is a white solid of substance

It is a white solid and is called as white

phosphorus pentoxide and is called as

white phosphorus and is called as white



On heating, the pure substance

forms a white solid which is called as

phosphorus pentoxide and phosphorus pentoxide

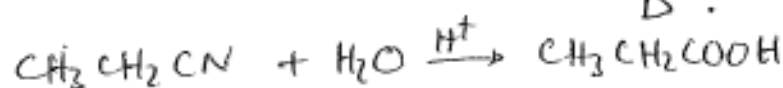
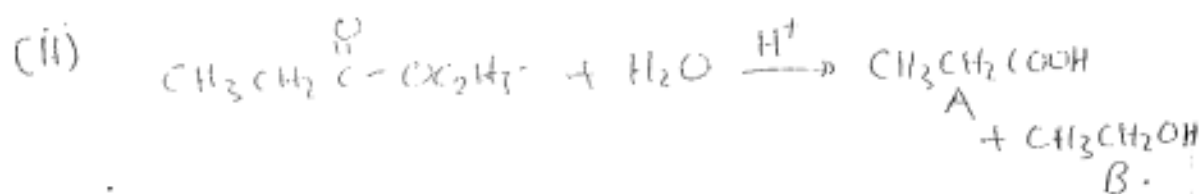
and phosphorus pentoxide is a white solid

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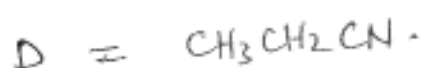
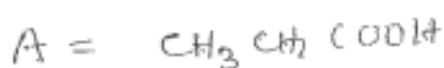
It is a white solid and is called as white



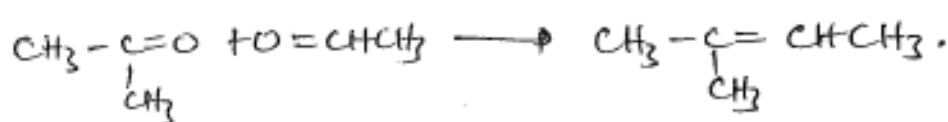
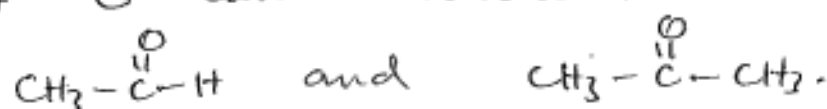
(a) (i) The reaction of alkyl halide with KCN increase one carbon and on hydrolysis it gave acid A. This shows that the s.f of ester E was $\text{CH}_3\text{CH}_2\overset{\text{O}}{\parallel}\text{C}-\text{OC}_2\text{H}_5$.
Name: Ethylpropanoate.



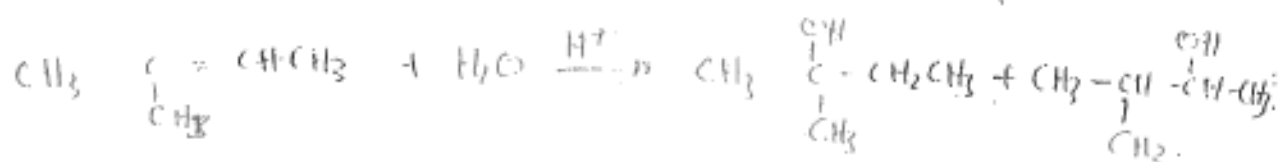
The s.f of



(b) From the products of ozonolysis the s.f of C can be deduced

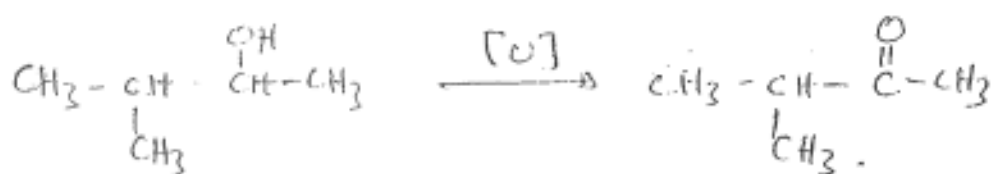


B can be formed by hydration of C.

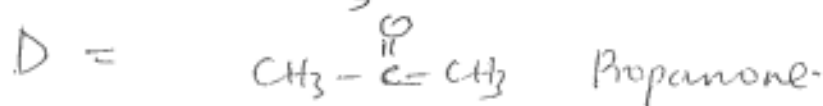
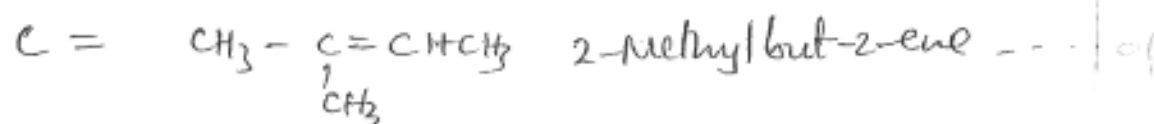
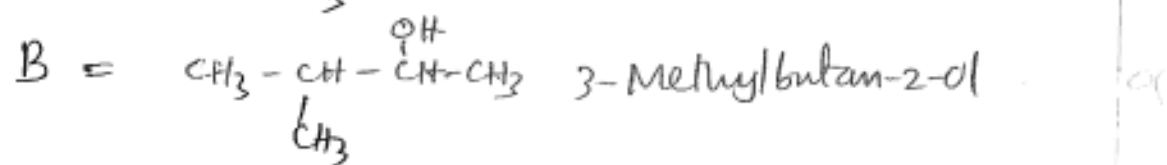
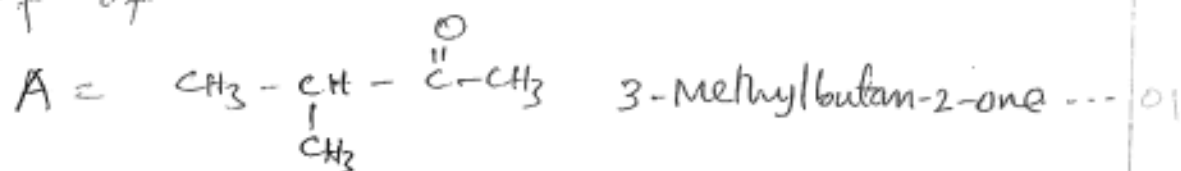


The correct s.f of B is $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \overset{\text{OH}}{\text{CH}} - \text{CH}_3$

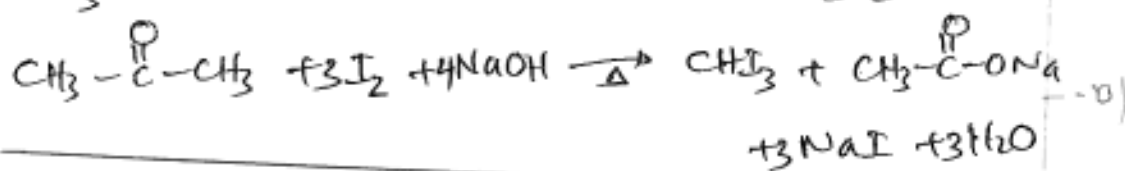
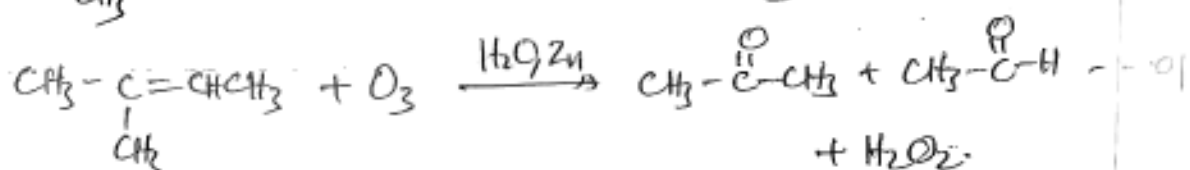
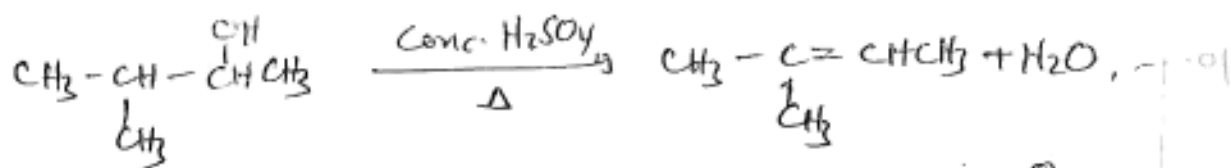
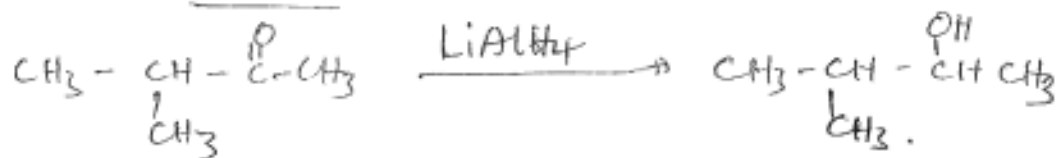
Oxidation of B gave A.



S.f of



Reactions:



(12)

6. (c)

